

Yihan Li

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Education

Sun Yat-sen University	Sep 2023 – Jul 2027 (Expected)
<i>B.A. in English Language & Literature, School of Foreign Languages</i>	<i>Major GPA: 3.8/4.0</i>
<i>Minor in Intelligent Unmanned Systems</i>	<i>GPA: 4.0/4.0</i>
<i>Minor in Computer Science & Technology</i>	<i>GPA: 3.8/4.0</i>

Publications

Anticipating Contact for Visuo-Tactile Diffusion Policies	Under Review
Yating Feng, Yuhang Zheng, Lixin Xu, Yihan Li , Jiaju Yin, Renjing Xu	
Goldsmith: Gold-Loss-Guided Prompt Training for Agentic Annotation	Under Review
Yihan Li , Hanyi Zhang, Xiaoxi Jiang, Man Guo	
Unordered Landmark Visual Navigation	ECCV 2026
Hao Ren, Junzhe Zhu, Yihan Li , Zetong Bi, Le Zheng, Zhi Li, Yiqing Yuan, Zhaoliang Wan, Lu Qi, Hui Cheng	
OmniTrav: A Dataset and Benchmark for 360° Vision-based Traversability via Foundation Models	IROS 2026
Yihan Li , Hao Ren, Zhenglan Jiang, Junzhe Zhu, Hui Cheng	
RAPID Hand: A Robust, Affordable, Perception-Integrated, Dexterous Manipulation Platform for Generalist Robot Autonomy	NeurIPS 2025
Zhaoliang Wan, Zetong Bi, Zida Zhou, Hao Ren, Yiming Zeng, Yihan Li , Lu Qi, Xu Yang, Ming-Hsuan Yang, Hui Cheng	

Work Experience

HCL Lab (HKUST(GZ))	Jan 2026 – Present
<i>Research Assistant</i>	<i>Prof. Renjing Xu</i>
- Researching tactile sensors and vision models for dexterous manipulation, including multimodal perception pipelines for visuo-tactile policy learning. - Fine-tuning VLA models and developing VLA-based online RL workflows, with Real2Sim scene reproduction for reproducible evaluation.	
Yun Drone	Jun 2025 – Mar 2026
<i>UAV Development Intern</i>	
- Selected UAV propulsion components and participated in formation-flight development, navigation and localization, and flight-test workflow design. - Explored RL-based disturbance-resilient control and analyzed flight logs to support anti-disturbance tuning and system iteration.	
RAPID Lab (SYSU)	Nov 2024 – Jun 2025
<i>Research Assistant</i>	<i>Prof. Hui Cheng</i>
- Developed the dexterous-hand control stack, visuo-tactile temporal alignment, and MuJoCo Sim2Real validation workflow for RAPID Hand manipulation experiments. - Contributed to OmniTrav by building the C2T pipeline, mapping pinhole, fisheye, and panoramic RGB inputs into planner-ready 360-degree radial clearance bounds.	
RoboTech Association (SYSU)	Sep 2024 – Present
<i>President; Competition Team Captain</i>	
- Led 160+ members and competition teams, coordinating training schedules, technical reviews, and cross-group collaboration. - Authored robotics onboarding tutorials across mechanical design, hardware, embedded control, and algorithms.	

Projects

Venom VNV: ROS 2 Unified Robotics Platform

Mar 2025 – Jul 2025

Project Lead; ROS 2 Platform Developer

- Architected a ROS 2 Humble platform spanning drivers, perception, localization, planning, mission control, simulation, and interface specifications for UGV/UAV/USV scenarios.
- Integrated Livox MID360, Hikvision cameras, AgileX chassis/arm/PX4 drivers with Point-LIO/Fast-LIO, Nav2 TEB, Ego Planner Swarm, YOLO detection, auto-aim tracking, and waypoint task management.

RAPID Hand: Perception-Integrated Dexterous Manipulation Platform

Nov 2024 – Jun 2025

Research Assistant

- Implemented drivers, visuo-tactile fusion, MuJoCo Sim2Real training, and Diffusion Policy tasks with 50/50 success.
- Integrated 1 kHz hand control, wrist vision, fingertip tactile sensing, and joint proprioception under 7 ms synchronization latency.

OmniTrav: 360° Vision-based Traversability Benchmark

Jun 2025 – Mar 2026

First Author

- Proposed C2T RGB traversability and built a 4.7k-scene, 250k-frame, 577 GB automated foundation-model dataset.
- Evaluated pinhole, fisheye, and panoramic cameras; achieved panoramic obstacle detection F1 of 1.0 and pinhole median error of 0.27 m.

GenreShift and Rosetta: LLM Systems for Linguistic Research

Oct 2024 – Nov 2025

Project Leader

- Built GenreShift for genre and discipline analysis, extracting quantitative linguistic features such as TTR, MTLT, academic vocabulary, and cross-genre comparisons for LoRA/MoE-based genre transformation.
- Developed Rosetta as an agentic annotation pipeline with concept bootstrap, prompt optimization, embedding-based top-k retrieval, LLM judge routing, review feedback, and Prodigy-compatible JSONL export.

Awards

China Robot and Artificial Intelligence Competition

2025

National First Prize (Team Captain)

National Intelligent Unmanned Systems Application Challenge

2025

National Champion (Team Captain)

National Undergraduate Electronics Design Contest

2025

National Second Prize (Team Captain)

National Undergraduate Robotics Competition - RoboMaster University League

2025

National Second Prize (Vision Group Leader)

Skills

Programming: ROS1/2, Python, C/C++, MATLAB, STM32/Arduino.

Robotics/ML: Visual navigation, 3D perception, dexterous manipulation, MuJoCo, TensorRT, OpenVINO, LLM annotation systems.

Engineering and Languages: Fusion 360, SolidWorks, EDA design; AOPA Multi-rotor VLOS Pilot; Mandarin (native), English (TEM-4), French.